

1. Surface Discovery

- BFS Web Crawler
 - Form / Link Ingestion
 - Query Parameter Parser
 - Max Depth Bounded (D_{max})

- OpenAPI / Swagger
 - v2.0 / v3.0 Parser
- Endpoint Schema Seeding
- HTTP Verb Registration

2. Scanner Engine (14)

Recon: Nmap, Nikto, DirBrute
Web: SQLi, XSS, CORS, Misconfig

API/Auth: APIAbuse, Auth, JWT
Exploit: CmdInj, SSRF, PathTrav

ThreadPool ($k = 8$ Workers)
Differential Baseline Cache

3. Intelligence Layer

- Compound Deduplication
- Key: host+port+URI+CWE
 - Multi-alert merging (-39.5%)

- Enrichment & Compliance
- NIST NVD & CIRCL CVEs
 - OWASP '21, CWE, PCI-DSS

4. Policy & Delivery

- CI/CD Policy Gating
- Native exit code (--fail-on)
 - Set-Theoretic Delta Diffing

- Multi-Channel Reporting
- Real-Time SSE Dashboard
 - PDF / HTML & Webhooks